Rajeev Yasarla

Contact Information

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Research Profile

Computer vision and machine learning researcher developing **predictive visual intelligence for robust embodied agents**. My work connects uncertainty-aware and data-efficient learning, temporal and 3D/4D scene understanding, vision–language–action models, and latent world models to prediction, planning, and control under distribution shift. At Qualcomm AI Research, I lead projects in end-to-end autonomous driving, world-action modeling, multimodal learning, and dynamic 3D perception. My research has appeared in NeurIPS, CVPR, ICCV, ECCV, ICML, WACV, IEEE TIP, and IEEE TPAMI and received the **Best Paper Award at IEEE ICIP 2021**.

Research Impact

- Google Scholar: **3,208 citations**, **h-index 19**, and **i10-index 20** (September 2026).
- Highly cited contributions include *TransWeather* (819 citations), uncertainty-guided image deraining (462), *JHU-CROWD++* (431), domain-adaptive object detection (284), and *Syn2Real* (269), as of September 2026.
- Recent first-author work includes *RoCA* (ICML 2026), *Generative Scenario Rollouts* (NeurIPS 2026), and *MAPLE* (NeurIPS 2026); the latter two improve closed-loop driving through language-conditioned and reactive multi-agent latent rollouts.
- **Translation and broader impact:** end-to-end driving and VLA demonstrations at NeurIPS 2025 and CES 2026, **five granted U.S. patents**, and four pending applications; robust perception and data-efficient medical imaging support safer mobility and assistive applications.

Research Areas

Robust and trustworthy computer vision; embodied AI; vision-language-action and world models; data-efficient and self-supervised learning; 3D/4D and dynamic scene understanding; robotics and autonomous driving.

Education

Johns Hopkins University, Baltimore, MD, USA Sep 2018 – May 2022
Ph.D. in Electrical Engineering
Advisor: Prof. Vishal M. Patel
Thesis: Deep Learning-Based Approaches for Image Restoration

Indian Institute of Technology Madras, Chennai, India Jul 2011 – May 2016
B.Tech. and M.Tech. in Electrical Engineering
Advisor: Prof. A. N. Rajagopalan
Thesis: Pedestrian Detection and Tracking for Driver Assistance

Research Experience

Senior Machine Learning Researcher, Qualcomm AI Research Nov 2022 – Present

- Lead research on end-to-end autonomous driving and embodied agents, including *RoCA*, *GeRo*, and *MAPLE* for cross-domain adaptation, future generation, and reactive multi-agent learning.
- On Bench2Drive, GeRo raises driving score by 15.7 points and success rate by 26.2 points; MAPLE raises driving score by over 25% using latent multi-agent interaction.
- Develop world-action models that map multi-view video into compact map, agent, and goal states, learn action-conditioned latent dynamics, and connect predicted states to policy learning.

- Led dynamic 3D perception research spanning video depth estimation, multi-camera 3D object detection, and multimodal model distillation, resulting in publications at ICCV, ECCV, and CVPR and multiple patent applications.
- Presented end-to-end autonomous driving and VLA research demonstrations at NeurIPS 2025 and CES 2026.

Computer Vision Research Scientist, AIBEE May 2022 – Oct 2022

- Conducted research on single-view joint object detection and tracking for visual perception systems.

Ph.D. Researcher, Johns Hopkins University Aug 2018 – Mar 2022

- Developed uncertainty-aware, semi-supervised, and self-supervised methods for image restoration and robust perception under adverse weather, atmospheric turbulence, blur, noise, and low light.
- Contributed to domain-adaptive object detection, crowd understanding, salient object detection, face enhancement, and data-efficient medical image analysis.

Research Intern, AIBEE May 2021 – Aug 2021

- Developed methods for unsupervised salient object detection, including pseudo-label generation and task-specific data augmentation.

Research Intern, Microsoft May 2020 – Aug 2020

- Developed neural methods for face enhancement under noise, blur, and low-light degradations.
- Contributed to patented technology: US Patent 11,526,972.

Research Assistant, Indian Institute of Science Sep 2017 – Jun 2018

- Conducted research on deep learning methods for joint image denoising and deblurring.

Master's Thesis Researcher, IIT Madras Jun 2015 – May 2016

- Developed computer vision algorithms for driver-assistance pedestrian and object detection.

Research Highlights

- **Predictive embodied intelligence.** Developed VLA and driving methods for cross-domain adaptation, language-conditioned future generation, and reactive multi-agent learning: *RoCA*, *GeRo*, *MAPLE*, and multimodal-LLM distillation.
- **Temporal and 3D/4D visual understanding.** Developed memory, prediction, state-space, and generative methods for video depth, multi-camera 3D detection, and text-to-3D: *MAMo*, *FutureDepth*, *DySS*, and *HexaGen3D*.
- **Robust and data-efficient learning.** Introduced uncertainty-aware and Gaussian-process methods for restoration, adverse-weather perception, domain adaptation, crowd understanding, and medical imaging. This work includes the IEEE ICIP 2021 Best Paper and several papers with more than 250 citations each.

Teaching And Mentoring

Introduction to Digital Signal Processing, Johns Hopkins University 2019 – 2021
Teaching Assistant; Course Instructor: Prof. Vishal M. Patel

Data Structures and Algorithms, IIT Madras Spring 2016
Teaching Assistant; Course Instructor: Prof. G. Venkatesh

Basic Electrical Engineering, IIT Madras Fall 2015
Teaching Assistant; Course Instructor: Prof. Ananth Krishnan

Research Mentoring and Supervision, Qualcomm AI Research 2022 – Present
Mentor graduate researchers and interns from problem formulation through publication, including external Ph.D. students through the Qualcomm Innovation Fellowship.

Awards And Honors

- **Best Paper Honorable Mention Award**, Wearables AI Workshop and Safe and Defensive Autonomous Driving Workshop, ECCV 2026.
- **Best Paper Award**, IEEE International Conference on Image Processing (IEEE ICIP), 2021.
- Ranked in the top 0.4% among nearly one million students in IIT-JEE 2011.
- First Prize, National Fox Hunt Event, Shaastra 2013, technical festival of IIT Madras.

Fellowships And Research Support

- Student Travel Grant, ICCV 2019, awarded by UIUC in association with the ICCV organizing committee.
- Merit-cum-Means Scholarship, Indian Institute of Technology Madras.
- Fully funded doctoral research support, Johns Hopkins University, 2018–2022.

Service And Leadership

Workshop Organizer: Vision-based Assistants in the Real World (VAR) Workshop, CVPR 2026; collaborating with academic and industry organizers to build a forum on visual assistants in real-world settings.
Area Chair: IEEE Winter Conference on Applications of Computer Vision (WACV), 2027.
Reviewer: NeurIPS, ICLR, CVPR, ICCV, ECCV, WACV, IEEE TIP, IJCV, AVSS, Pattern Recognition Letters, and IEEE Signal Processing Magazine.
Leadership and Outreach: Lead Coordinator, Shaastra Circuit Design Challenge, IIT Madras, 2014.

Patents

1. *Test-Time Self-Supervised Guidance for Diffusion Models*, U.S. Patent 12,738,035, 2026. Co-inventor.
2. *3D Reconstruction Using 3D Triplane Representation*, U.S. Patent 12,700,181, granted August 4, 2026. Co-inventor.
3. *Depth Estimation Based on Feature Reconstruction with Adaptive Masking and Motion Prediction*, U.S. Patent 12,694,550, granted July 28, 2026. Lead-inventor.
4. *Unified Simultaneous Optical Flow and Depth Estimation*, U.S. Patent 12,561,822, granted February 24, 2026. Co-inventor.
5. *Simultaneously Correcting Image Degradations of Multiple Types in an Image of a Face.* U.S. Patent 11,526,972, 2022. Co-inventor.
6. *Language-Grounded Vehicle Path Planning*, U.S. Patent App. 18/934,765, 2026. Lead-inventor.
7. *Three-Dimensional Object Detection Using State-Space Spatiotemporal Learning and Dynamic Queries*, U.S. Patent App. 18/882,586, 2026. Lead-inventor.
8. *Memory-Oriented Gaussian Process-Based Multi-Object Tracking.* U.S. Patent App. 18/339,408, 2024. Lead-inventor.
9. *Monocular Image Depth Estimation with Attention*, U.S. Patent App. 18/538,869, 2024. Lead-inventor.

Preprints

1. S Mahajan, S Kadambi, H Le, **R Yasarla**, A Bhattacharyya, M Hayat, et al., *Do-Undo Bench: Reversibility for Action Understanding in Image Generation*, arXiv, 2025.

Conference Publications

1. **R Yasarla**, D Hegde, S Han, H Cheng, Y Shi, M Sadeghigooghari, S Mahajan, A Bhattacharyya, L Liu, R Garrepalli, T Svantesson, F Porikli, H Cai, *Generative Scenario Rollouts for End-to-End Autonomous Driving*, **NeurIPS**, 2026. Accepted.
2. **R Yasarla**, D Hegde, H Cheng, S Han, Y Shi, M Sadeghigooghari, H Ackermann, L Liu, P Desai, F Porikli, M Ghavamzadeh, H Cai, *MAPLE: Latent Multi-Agent Play for End-to-End Autonomous Driving*, **NeurIPS**, 2026. Accepted.
3. **R Yasarla**, S Han, H Cheng, L Liu, S Mahajan, A Bhattacharyya, Y Shi, R Garrepalli, H Cai, F Porikli, *RoCA: Robust Cross-Domain End-to-End Autonomous Driving*, **ICML**, 2026.
4. A Bhattacharyya, S Mahajan, S Haresh, **R Yasarla**, R Pourreza, L Liu, et al., *Streaming Interventions: Can Video Large Language Models Correct Mistakes as They Occur?*, **ECCV Workshops (Wearables AI)**, 2026.
5. D Hegde*, **R Yasarla***, H Cai, S Han, A Bhattacharyya, S Mahajan, L Liu, R Garrepalli, V Patel, F Porikli, *Distilling Multi-modal Large Language Models for Autonomous Driving*, **CVPR**, 2025. *Equal contribution.
6. **R Yasarla**, S Han, H Cai, F Porikli, *DySS: Dynamic Queries and State-Space Learning for Efficient 3D Object Detection from Multi-Camera Videos*, **CVPR Workshops**, 2025.
7. A Mercier, R Nakhli, M Reddy, **R Yasarla**, H Cai, F Porikli, G Berger, *HexaGen3D: StableDiffusion Is One Step Away from Fast and Diverse Text-to-3D Generation*, **WACV**, 2025.
8. **R Yasarla**, MK Singh, H Cai, Y Shi, J Jeong, Y Zhu, S Han, R Garrepalli, F Porikli, *FutureDepth: Learning to Predict the Future Improves Video Depth Estimation*, **ECCV**, 2024.
9. **R Yasarla**, J.M.J. Valanarasu, V.M. Patel, *Self-Supervised Denoising Transformer with Gaussian Process*, **WACV**, 2024.
10. **R Yasarla**, R Weng, W Choi, V Patel, A Sadeghian, *3SD: Self-Supervised Saliency Detection with No Labels*, **WACV**, 2024.
11. **R Yasarla**, H Cai, J Jeong, Y Shi, R Garrepalli, F Porikli, *MAMo: Leveraging Memory and Attention for Monocular Video Depth Estimation*, **ICCV**, 2023.
12. **R Yasarla**, C.E. Priebe, V.M. Patel, *ART-SS: An Adaptive Rejection Technique for Semi-Supervised Restoration for Adverse Weather-Affected Images*, **ECCV**, 2022.
13. J.M.J. Valanarasu, **R Yasarla**, V.M. Patel, *TransWeather: Transformer-Based Restoration of Images Degraded by Adverse Weather Conditions*, **CVPR**, 2022.
14. **R Yasarla**, V.A. Sindagi, V.M. Patel, *Unsupervised Restoration of Weather-Affected Images Using Deep Gaussian Process-Based CycleGAN*, **ICPR**, 2022. **Oral presentation**.
15. N.G. Nair, **R Yasarla**, V.M. Patel, *NBD-GAP: Non-Blind Image Deblurring without Clean Target Images*, **IEEE ICIP**, 2022.
16. **R Yasarla**, H.V. Joze, V.M. Patel, *Network Architecture Search for Face Enhancement*, **EUSIPCO**, 2022. **Oral presentation**.
17. **R Yasarla**, V.M. Patel, *Learning to Restore Images Degraded by Atmospheric Turbulence Using Uncertainty*, **IEEE ICIP**, 2021. **Best Paper Award**.
18. **R Yasarla**, V.A. Sindagi, V.M. Patel, *Syn2Real Transfer Learning for Image Deraining Using Gaussian Processes*, **CVPR**, 2020. **Oral presentation**.
19. V.A. Sindagi*, P Oza*, **R Yasarla**, V.M. Patel, *Prior-Based Domain Adaptive Object Detection for Adverse Weather Conditions*, **ECCV**, 2020.
20. V.A. Sindagi, **R Yasarla**, D S Babu, R. V. Babu, V.M. Patel, *Learning to Count in the Crowd from Limited Labeled Data*, **ECCV**, 2020.

21. **R Yasarla**, V.M. Patel, *Uncertainty Guided Multi-Scale Residual Learning Using a Cycle Spinning CNN for Single Image Deraining*, **CVPR**, 2019.
22. V.A. Sindagi, **R Yasarla**, V.M. Patel, *Pushing the Frontiers of Unconstrained Crowd Counting: New Dataset and Benchmark Method*, **ICCV**, 2019.
23. N.J. Myers, **R Yasarla**, *Analysis of 8-Point Constellations: Optimality and Recovery*, **IEEE ICSC**, 2016.

Journal Publications

1. **R Yasarla**, V.M. Patel, *CNN-Based Restoration of a Single Face Image Degraded by Atmospheric Turbulence*, **IEEE TBIOM**, 2022.
2. **R Yasarla**, V.A. Sindagi, V.M. Patel, *Semi-Supervised Image Deraining Using Gaussian Processes*, **IEEE TIP**, 2021.
3. V.A. Sindagi, **R Yasarla**, V.M. Patel, *JHU-CROWD++: Large-Scale Crowd Counting Dataset and a Benchmark Method*, **IEEE TPAMI**, 2020.
4. **R Yasarla**, V.M. Patel, *Confidence Measure Guided Single Image Deraining*, **IEEE TIP**, 2020.
5. **R Yasarla**, F Perazzi, V.M. Patel, *Deblurring Face Images Using Uncertainty Guided Multi-Stream Semantic Networks*, **IEEE TIP**, 2020.
6. **R Yasarla***, J.M.J. Valanarasu*, V.M. Patel, *Exploring Overcomplete Representations for Single Image Deraining Using CNNs*, **IEEE JSTSP**, 2020.
7. J.M.J. Valanarasu, **R Yasarla**, P Wang, I Hacihaliloglu, V.M. Patel, *Learning to Segment Brain Anatomy from 2D Ultrasound with Less Data*, **IEEE JSTSP**, 2020.
8. P Guo, P Wang, **R Yasarla**, J Zhou, V.M. Patel, S Jiang, *Anatomic and Molecular MR Image Synthesis Using Confidence-Guided CNNs*, **IEEE TMI**, 2020.

Book Chapter

1. **R Yasarla**, F Perazzi, V.M. Patel, *Deblurring Face Images Using Deep Networks*, in *Deep Learning-Based Face Analytics*, Advances in Computer Vision and Pattern Recognition, 2021.

Technical Skills

Programming: Python, C, C++
Machine Learning and Vision: PyTorch, TensorFlow, OpenCV, MATLAB
Tools: LaTeX, research prototyping, deep learning model development, computer vision system evaluation

Selected Talks And Demos

- Live demonstrations of end-to-end autonomous driving and vision–language–action systems, NeurIPS 2025 and CES 2026.
- Oral presentation, *Syn2Real Transfer Learning for Image Deraining Using Gaussian Processes*, CVPR 2020.
- Oral presentations on self-supervised restoration and face enhancement, ICPR 2022 and EUSIPCO 2022.

References

Prof. Dr. Vishal M. Patel, Johns Hopkins University — vpatel36@jhu.edu
Prof. Dr. Rama Chellappa, Johns Hopkins University — rchella4@jhu.edu
Prof. Dr. Carey E. Priebe, Johns Hopkins University — cep@jhu.edu